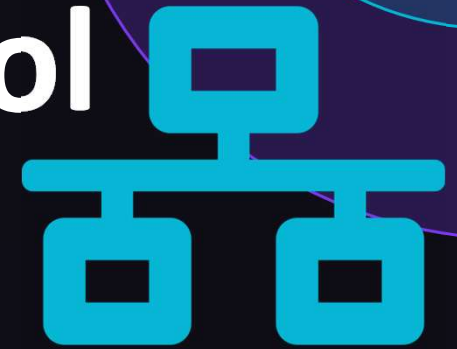


DATA THINKERS

Introduction to Model Context Protocol

Building & Implementing MCP using n8n



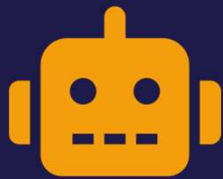
MCP

- What is MCP
- Why MCP is needed
- Tools & AI agents
- MCP Architecture
- How MCP works internally



The Real Need

The Senior Employee



- ✓ Very intelligent
- ✓ Understands instructions
- ✓ Can analyze problems
- ✓ Can make decisions



BUT...

Needs Access To



Database



Email systems



Calendar apps



Internal software



Without tools, even the most intelligent employee can only talk — not act.



Modern AI Systems Are Changing

BEFORE — Earlier AI

- AI models mostly generated answers
- You ask a question → AI responds
- That was the entire interaction
- Limited to text generation only

VS

NOW — Modern AI Agents

Use tools

Search databases

Access APIs

Work with applications

Read files

Perform actions automatically



Agentic AI

AI is no longer just generating text — it can actually interact with systems and perform real tasks.

This is where Agentic AI becomes really powerful.

👉 *AI can actually interact with systems — not just talk about them.*



What are Tools?

Tools are functionalities that AI agents can use to perform real-world tasks.



Calculator

Math & computations



Database

Read/write data



Email API

Send messages



API Tool

Connect services



Google Drive

Access files



Code Runner

Execute scripts



AI agents use tools to perform REAL actions — not just generate text!

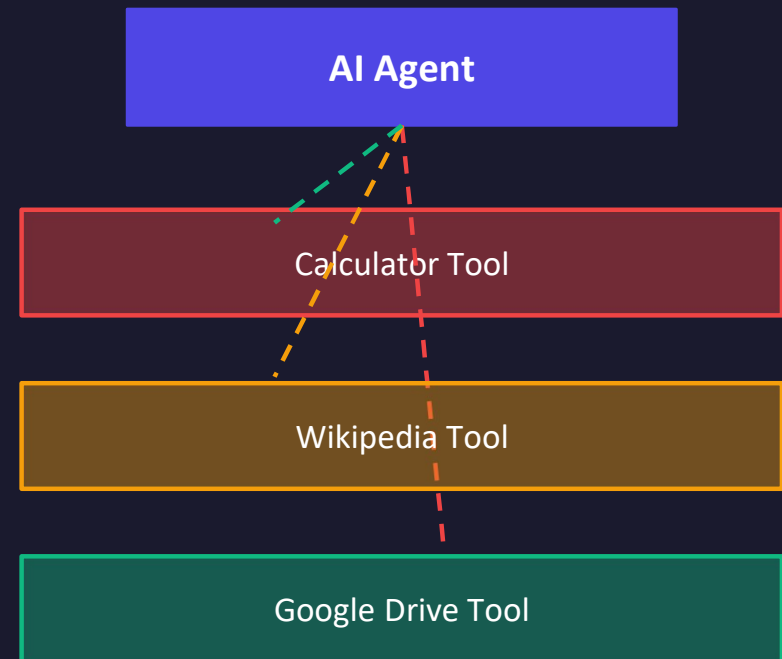


The Problem: Growing Complexity

As systems grow...

- Multiple agents need same tools
- Multiple workflows share integrations
- Different apps need same resources
- Every tool has different configs
- Repeated reconnections required
- Management becomes messy

System WITHOUT MCP



👉 *Tightly coupled tools → hard to scale, painful to maintain.*



What is MCP?

M

Model

C

Context

P

Protocol



Introduced by Anthropic — now one of the most important concepts in modern Agentic AI



"USB-C for AI applications"

Just as USB-C standardized device connections, MCP standardizes how AI systems connect and communicate with tools.



MCP provides a **STANDARDIZED** way for AI systems to connect with tools and external systems.



System WITH MCP



AI Agent → MCP Client → MCP Server → Tools



Reusability

Multiple agents can reuse the same tools



Shared Ecosystem

Multiple apps share one set of integrations



Standardization

One protocol for all AI-to-tool communication



Scalability

Easily add agents without duplicating tools



How MCP Manages Tools

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MCP Server Exposes

- Tool name
- Tool description
- Capabilities
- Required parameters
- How responses are handled



AI Agent Now Knows

- ✓ Which tools are available
- ✓ What each tool does
- ✓ When a tool should be used
- ✓ What inputs are required
- ✓ How to handle responses

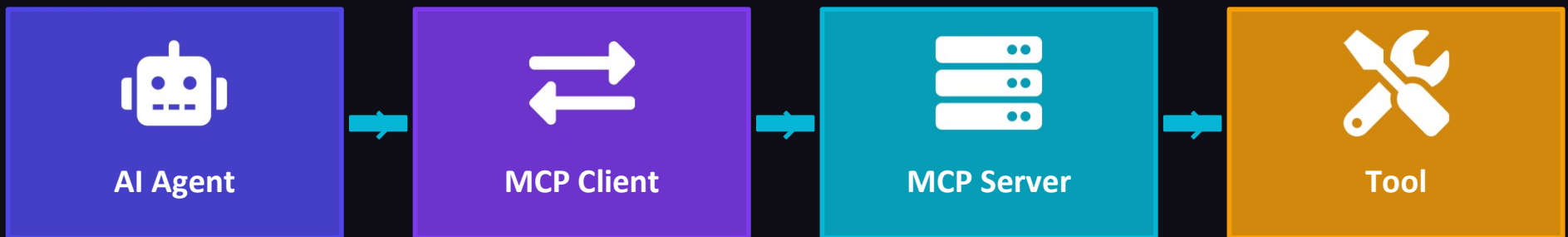
💬 User asks: "What is $12345 * 9876$?" → AI decides: calculator tool needed → MCP routes the call → Answer returned!



MCP Manages the Communication

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Then MCP manages communication between all parts:



The tool performs the actual task



The response comes back through MCP



AI generates the final answer for the user



Important:

MCP is NOT performing the task — MCP is managing and standardizing how tools are exposed and used.



Important Understanding

MCP is NOT the tool itself.

The tools still perform the actual work.



MCP IS the bridge between AI systems and tools.

MCP standardizes how AI systems connect and communicate with tools.

AI Systems



MCP
(The Bridge)



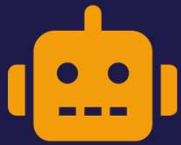
Tools

MCP simply standardizes how AI systems connect and communicate with tools.



MCP Architecture: 3 Core Components

MCP Host



The AI Application

- Claude Desktop
- VS Code
- AI applications
- n8n workflows

The main environment where the agent runs

MCP Client



The Connector

- Sends requests
- Receives responses
- Handles communication
- Bridges host & server

Communicates between host and MCP server

MCP Server



The Exposer

- Exposes Tools
- Exposes Resources
- Exposes Prompts
- Lists available tools

Tells clients which tools are available



Simple MCP Flow — Step by Step Example



User asks: "What is $12345 * 9876$?"

1

AI Agent receives the question

2

Agent decides: Calculator tool is needed

3

MCP Client connects to MCP Server

4

MCP Server exposes the calculator tool

5

Calculator tool performs the calculation

6

Response travels back through MCP Client

7

AI Agent generates final answer for user ✓



MCP: Key Takeaways

⚠ Important

MCP is NOT the tool itself.

MCP is the BRIDGE between AI systems and tools.

*The tools still perform the actual work.
MCP standardizes how AI systems connect and
communicate with them.*

✓ Benefits of MCP

- Standardization
- Reusability
- Easier integrations
- Shared tool ecosystems
- Better scalability

With MCP : AI Agent → MCP Client → MCP Server → Tools



Next: Building MCP inside n8n — now it all makes perfect sense!